



ASSIGNMENT 1

Develop an ML Model Using Linear Regression

OBJECTIVE

Build, train, evaluate and test a Machine Learning model using Linear Regression to predict a continuous numerical value.

PROJECT OVERVIEW

Suggested Project

House Price Prediction

Build a regression model that predicts house prices using property-related features.

INPUT FEATURES	TARGET
Area / Square Feet	House Price
Bedrooms	Continuous numerical value
Bathrooms	
Number of Floors	
Age of House	
Location / Other relevant features	

TECHNOLOGY STACK

Python	NumPy	Pandas
Matplotlib	Seaborn	Scikit-learn

PROJECT OUTCOME

By completing this assignment, you should be able to take a real-world dataset, prepare it, train a Linear Regression model, evaluate its performance, visualize results, and make predictions for new data.



ASSIGNMENT TASKS

Complete all tasks below and maintain your code, outputs, graphs and observations in your project notebook.

01	Understand the Business Problem	Identify the problem, required data, important factors affecting house prices, target variable and expected business outcome.
02	Collect / Download Dataset	Find a suitable House Price dataset with multiple input features and a numerical target variable.
03	Load Dataset Using Pandas	Read the dataset into a Pandas DataFrame and inspect the initial records.
04	Perform Data Understanding	Use head(), shape, info(), describe() and missing-value checks.
05	Perform Exploratory Data Analysis	Analyze distributions, correlations, relationships between features and target, and possible outliers.
06	Identify X and y	Select the independent variables X and the dependent / target variable y.
07	Data Preprocessing	Handle missing values, unnecessary columns, duplicate records, incorrect data types and categorical features.
08	Train / Test Split	Split the dataset into 80% training data and 20% testing data using train_test_split().
09	Build Linear Regression Model	Create a Scikit-learn LinearRegression() model.
10	Train the Model	Train the model using the training data so it can learn the relationship between features and house price.
11	Make Predictions	Use the trained model to predict house prices for the test dataset.
12	Evaluate the Model	Calculate MAE, MSE, RMSE and R ² Score and interpret the results.
13	Actual vs Predicted	Create a clear comparison between actual house prices and predicted house prices.
14	Visualize Results	Create suitable charts such as Actual vs Predicted, scatter plots and other relevant visualizations.
15	Test New / Unseen Data	Provide a new house record that was not used during training/testing and predict its price.
16	Write the Conclusion	Summarize model performance, important observations, limitations and possible improvements.



TECHNICAL REQUIREMENTS

1. Dataset Understanding

- head() - display the first few records.
- shape - identify number of rows and columns.
- info() - inspect columns, data types and non-null values.
- describe() - view statistical summaries.
- Missing values - identify and handle null values.

2. Exploratory Data Analysis

Analysis	Recommended Visualization
Feature distributions	Histogram
Feature vs House Price	Scatter Plot
Potential outliers	Box Plot
Numerical relationships	Correlation Heatmap
Model performance	Actual vs Predicted Plot

3. Model Evaluation

Metric	Meaning	Interpretation
MAE	Mean Absolute Error	Lower is better
MSE	Mean Squared Error	Lower is better
RMSE	Root Mean Squared Error	Lower is better
R ² Score	Explained variance / goodness of fit	Higher is better

4. New Data Prediction

Feature	Sample Value
Area	2000 sq.ft
Bedrooms	3
Bathrooms	2
Floors	2
Age	5 years

Prediction Flow

New House Details → Trained ML Model → Predicted House Price



SUBMISSION & LEARNING OUTCOMES

Expected Learning Outcomes

01	Understand a real-world Regression problem.
02	Load and analyze datasets using Pandas.
03	Perform EDA and basic data preprocessing.
04	Identify independent variables and the target variable.
05	Split data into training and testing sets.
06	Build and train a Linear Regression model.
07	Make predictions on unseen data.
08	Calculate and interpret MAE, MSE, RMSE and R ² Score.
09	Visualize model performance.
10	Explain the complete ML workflow confidently.

Submission Checklist

01	Dataset collected / downloaded	STATUS
02	Dataset loaded using Pandas	STATUS
03	Data understanding completed	STATUS
04	Missing values checked and handled	STATUS
05	EDA and visualizations completed	STATUS
06	X and y identified	STATUS
07	Preprocessing completed	STATUS
08	80 / 20 train-test split completed	STATUS
09	Linear Regression model created	STATUS
10	Model trained and predictions generated	STATUS
11	MAE, MSE, RMSE and R ² Score calculated	STATUS
12	Actual vs Predicted comparison created	STATUS
13	New / unseen data tested	STATUS
14	Conclusion written	STATUS



STUDENT CHALLENGE

Do not just run the code. Understand every step and be ready to explain:

Why are we doing this step? What happens internally? How does it help the ML model?

Goal: Explain your project confidently from Business Problem → Data → Model → Prediction → Evaluation.

Assignment Standard: Submit clean code, meaningful visualizations, metric results, observations and a clear conclusion. Your notebook should be understandable to another developer or trainer.